WORKED SOLUTION

STEP I 2005, Question 13

Probability and Statistics • Symmetric Distributions

STEP I 2005, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q13

The key idea

We are told nothing about the shape of the distribution except two cumulative probabilities and symmetry, so everything has to be built out of those. Symmetry about μ gives two more facts for free: half the probability lies on each side of μ, and the tail below $\mu - k\sigma$ matches the tail above $\mu + k\sigma$. In part (ii) the trick is to notice that every volume mentioned (495, 500, 505 ml) sits at μ, $\mu \pm \frac{1}{2}\sigma$ or $\mu + \sigma$ for one type of milk or the other.

Solution

Throughout we treat X as continuous (it models a volume), so a single value has probability zero and it doesn't matter whether we write $<$ or $\leq$.

Symmetry about μ gives $\mathrm{P}(X \leq \mu) = \frac{1}{2}$ and $\mathrm{P}(X \leq \mu - k\sigma) = \mathrm{P}(X \geq \mu + k\sigma)$ for any k. In particular

$$\mathrm{P}(X \leq \mu - \tfrac{1}{2}\sigma) = 1 - b, \qquad \mathrm{P}(X \leq \mu - \sigma) = 1 - a.$$

It helps to record the probability of each strip, measured in units of σ from μ:



For example, $\mathrm{P}(\mu < X \leq \mu + \frac{1}{2}\sigma) = b - \frac{1}{2}$ and $\mathrm{P}(\mu + \frac{1}{2}\sigma < X \leq \mu + \sigma) = a - b$; the left-hand strips follow by symmetry.

Part (i)

Subtracting cumulative probabilities,

$$\mathrm{P}\big(\mu - \tfrac{1}{2}\sigma \leq X \leq \mu + \sigma\big) = \mathrm{P}(X \leq \mu + \sigma) - \mathrm{P}\big(X < \mu - \tfrac{1}{2}\sigma\big) = a - (1 - b) = a + b - 1.$$

For the conditional probability, the event in the numerator is $\{\mu - \frac{1}{2}\sigma \le X \le \mu + \frac{1}{2}\sigma\}$, which has probability $b - (1-b) = 2b - 1$, while $\mathrm{P}(X \ge \mu - \frac{1}{2}\sigma) = 1 - (1-b) = b$. Hence

$$\mathrm{P}\big(X \le \mu + \tfrac{1}{2}\sigma \mid X \ge \mu - \tfrac{1}{2}\sigma\big) = \frac{2b-1}{b}.$$

Part (ii)(a)

Let S be the event that the carton is skimmed and F that it is full-fat, so $\mathrm{P}(S) = 0.6$ and $\mathrm{P}(F) = 0.4$. Let V be the volume in ml.

Ask yourself Where do 500 and 505 ml sit for each type of milk, in units of σ?

For skimmed milk ($\mu = 500$, $\sigma = 10$), 500 is μ and 505 is $\mu + \frac{1}{2}\sigma$. For full-fat milk ($\mu = 495$, $\sigma = 10$), 500 is $\mu + \frac{1}{2}\sigma$ and 505 is $\mu + \sigma$. Reading off the strips:

$$\begin{aligned} &\mathrm{P}(V < 505 \mid S) = b, && \mathrm{P}(500 < V < 505 \mid S) = b - \tfrac{1}{2}, \\ &\mathrm{P}(V < 505 \mid F) = a, && \mathrm{P}(500 < V < 505 \mid F) = a - b. \end{aligned}$$

By the law of total probability,

$$\begin{aligned} \mathrm{P}(V < 505) &= 0.6b + 0.4a, \\ \mathrm{P}(500 < V < 505) &= 0.6\big(b - \tfrac{1}{2}\big) + 0.4(a-b) = 0.4a + 0.2b - 0.3. \end{aligned}$$

The event "more than 500 ml and less than 505 ml" is the intersection we want, so

$$\mathrm{P}(V > 500 \mid V < 505) = \frac{0.4a + 0.2b - 0.3}{0.4a + 0.6b} = \frac{4a + 2b - 3}{4a + 6b}.$$

Part (ii)(b)

With a very large number of randomly chosen cartons, the observed proportions estimate the corresponding probabilities. The first piece of information says $\mathrm{P}(V \le 505) = 0.7$, which by part (a)'s calculation is

$$0.6b + 0.4a = 0.7, \quad \text{i.e.} \quad 4a + 6b = 7. \tag{1}$$

Ask yourself What is $\mathrm{P}(V \ge 495)$ for each type of milk?

For skimmed milk, 495 is $\mu - \frac{1}{2}\sigma$, so $\mathrm{P}(V \ge 495 \mid S) = b$ (this was the denominator in part (i)). For full-fat milk, 495 is μ, so $\mathrm{P}(V \ge 495 \mid F) = \frac{1}{2}$ by symmetry. By Bayes' theorem,

$$\mathrm{P}(F \mid V \ge 495) = \frac{0.4 \times \frac{1}{2}}{0.6b + 0.4 \times \frac{1}{2}} = \frac{0.2}{0.6b + 0.2}.$$

Setting this equal to $\frac{1}{3}$ gives $0.6b + 0.2 = 0.6$, so $b = \frac{2}{3}$. Then (1) gives $4a = 7 - 4 = 3$, so

$$a \approx \tfrac{3}{4}, \qquad b \approx \tfrac{2}{3}.$$

These are only estimates, because the observed proportions only approximate the true probabilities.

Checks

- The estimates are sensible: $\frac{1}{2} < b < a < 1$, as they must be, since $\mu < \mu + \frac{1}{2}\sigma < \mu + \sigma$.
- Substituting back: $0.6 \cdot \frac{2}{3} + 0.4 \cdot \frac{3}{4} = 0.4 + 0.3 = 0.7$, and $0.2/(0.4 + 0.2) = \frac{1}{3}$.
- For a Normal distribution $a \approx 0.841$ and $b \approx 0.691$; with these, part (i) gives 0.533 and 0.553. Simulating a (non-Normal) Laplace distribution with 10^5-plus samples matched all the formulae in parts (i) and (ii)(a); for instance, (ii)(a) gave 0.2518 against the formula's 0.2515.

Common slips

- Writing $\mathrm{P}(X \leq \mu - \frac{1}{2}\sigma) = b$ instead of $1 - b$: symmetry reflects the *tail*, not the cumulative probability.
- Forgetting that the two milk types have different means, so the same volume (say 500 ml) is a different number of standard deviations from the mean for each.
- Using $\mathrm{P}(V \geq 495) = b$ for both types; for full-fat milk 495 ml is the mean.

Beyond STEP

The point of the question is that you can get a long way with symmetry and the law of total probability without knowing the distribution at all; the mixture of two populations here is a simple *mixture model*, and part (ii)(b) is a baby version of estimating its parameters from data. A related distribution-free result is Chebyshev's inequality, which bounds $\mathrm{P}(|X - \mu| \geq k\sigma)$ using only the variance. See Blitzstein and Hwang, *Introduction to Probability*, or Grimmett and Welsh, *Probability: An Introduction*.

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